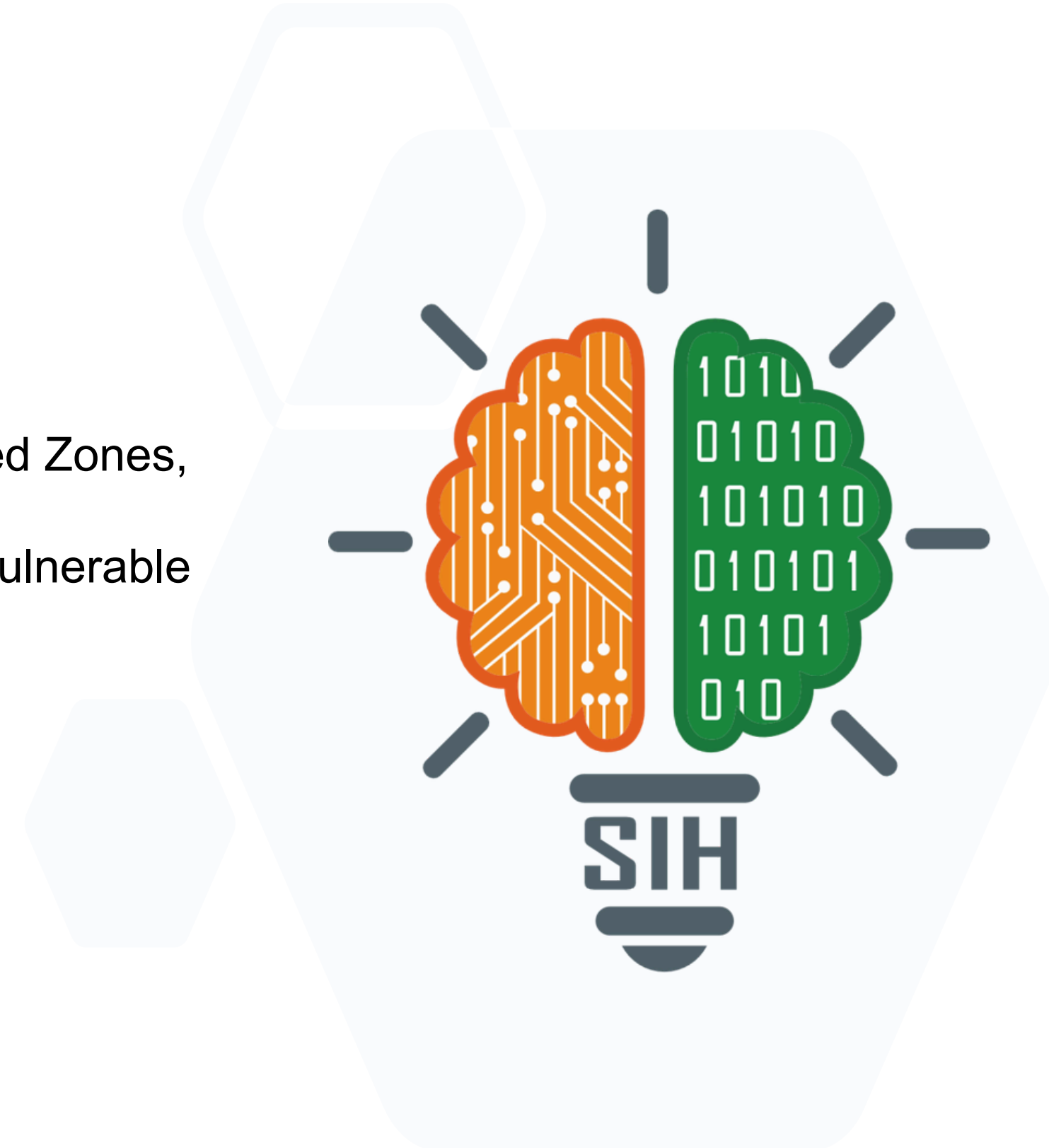


TITLE PAGE

- **Problem Statement ID** – SIH26191
- **Problem Statement Title-** Intelligent Identification of Hazard-Based Red Zones, Carrying Capacity Assessment, and Immediate Relocation Needs for Vulnerable Habitations
- **Theme-** Disaster Management
- **PS Category-** Software
- **Team ID-** BA8634
- **Team Name (Registered on portal)** - ByteX



Proposed Solution

A GIS-based platform that integrates hazard, population, vulnerability, terrain, infrastructure and historical disaster data to proactively identify vulnerable habitations and recommend relocation strategies.

How it addresses the problem



Multi-Hazard Analysis

- Combines flood, landslide, erosion, cloudburst and other hazard layers.
- Identifies habitation-level exposure and dynamically delineates high-risk Red Zones.



Vulnerability Assessment

- Evaluates population exposure, density, accessibility, historical impact and critical infrastructure.
- Generates an explainable vulnerability/risk score for each habitation.



Safe-Site & Carrying Capacity

- Identifies potential safer relocation sites.
- Evaluates safety, available capacity, infrastructure, accessibility and distance.



Relocation Prioritization

- Ranks vulnerable habitations into Immediate, Short and Medium-Term priorities.
- Recommends the most suitable feasible destination rather than simply the nearest location.

Innovation / Uniqueness

EXISTING APPROACH



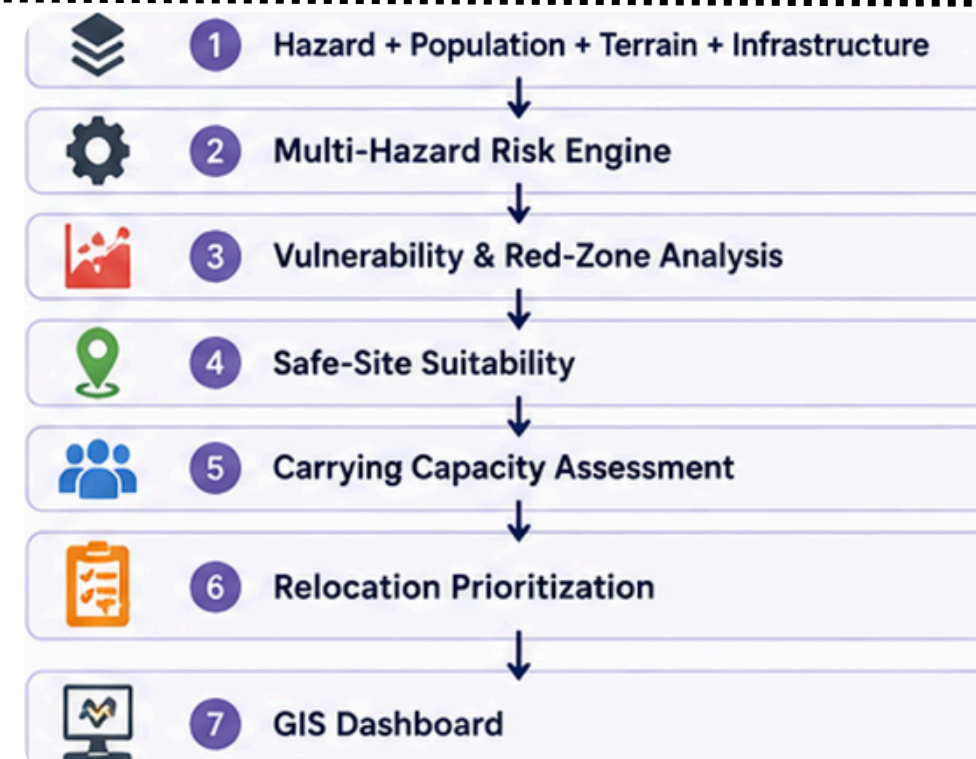
Hazard Map

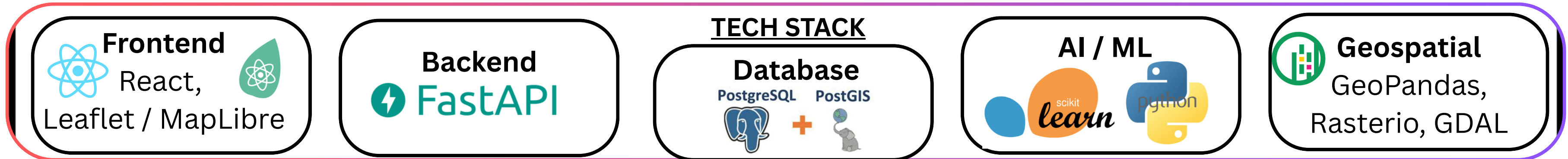
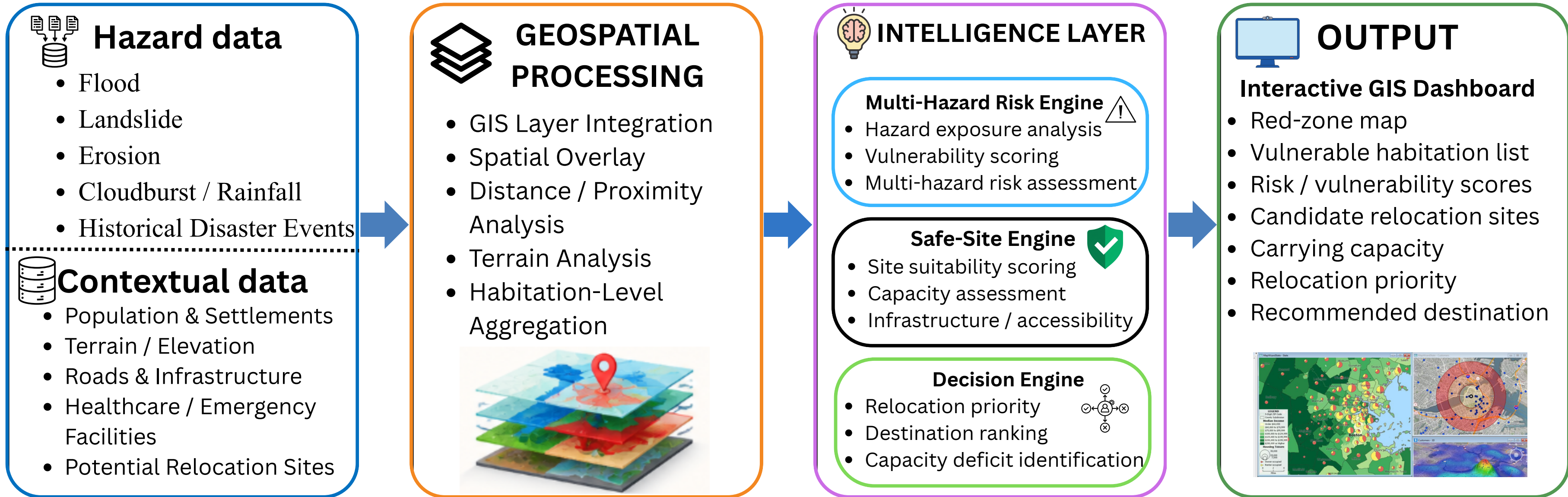
↓
Risk Information

Our Approach



Hazard → Vulnerability → Red Zone → Safe Site →
Capacity → Relocation Priority 





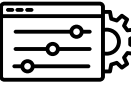
FEASIBILITY

-  Technically feasible with existing GIS, ML and open-source tools.
-  Required datasets are publicly available or accessible.
-  Software-based solution; no specialized hardware required.
-  Can be piloted in a selected disaster-prone district.

POTENTIAL CHALLENGES AND RISKS

-  Data heterogeneity and inconsistency.
-  Accuracy and reliability of risk scores.
-  Estimating carrying capacity of safe sites.
-  Data gaps in some regions or parameters.

STRATEGIES FOR OVERCOMING CHALLENGES

-  Standardized data preprocessing (uniform formats, CRS, quality checks).
-  Explainable and configurable risk scoring with expert-defined rules.
-  Multi-source validation and cross-verification of datasets.
-  Confidence indicators and uncertainty handling for recommendations.

VIABILITY



District Pilot

Test and validate in one district.



State-Level Deployment

Expand to multiple districts.



National Decision Support Platform

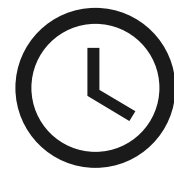
Scalable platform for nationwide use.



Impact on Target Audience



- Faster and informed decisions for disaster management authorities.



- Earlier risk identification and safer, planned relocation for vulnerable communities.



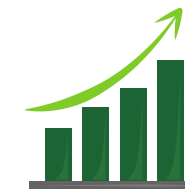
- Reduced exposure to hazards and better access to essential services.



Benefits of the Solution



- **Social:** Safer communities and improved quality of life.

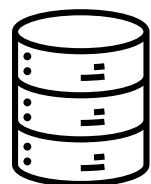


- **Economic:** Reduced losses and optimized use of resources.



- **Environmental:** Reduced impact on ecologically sensitive areas.

Overall Impact



Better Data
Integration



Accurate Analysis
and Planning



Effective and Timely
Relocation



Resilient and Safe
Communities

Official /
Government Sources

- Ministry of Home Affairs – Disaster Management
- National Disaster Management Authority (NDMA)
- ISRO / Bhuvan – Geospatial Resources
- Census of India – Population Data

<https://www.mha.gov.in/><https://ndma.gov.in/><https://bhuvan.nrsc.gov.in/><https://censusindia.gov.in/>Geospatial Data
& Tools

- OpenStreetMap – Mapping & Infrastructure Data
- GeoPandas – Spatial Data Processing
- GDAL – Raster & Vector Data Processing
- PostGIS – Spatial Database
- Scikit-learn – Machine Learning Library

<https://www.openstreetmap.org/><https://geopandas.org/><https://gdal.org/><https://postgis.net/><https://scikit-learn.org/>Research
Areas

- Multi-Hazard Risk Assessment
- GIS-Based Vulnerability Mapping
- Safe-Site Suitability & Carrying Capacity Estimation
- AI for Decision Support in Disaster Management

